TecNM Campus Ciudad Guzmán
Maestría en Ciencias Computacionales
Materia: Temas Selectos
Docente: Dr. Julio Alberto García Rodríguez

Class Project U3 – U4

Seminar: Machine Learning and Deep Learning for Time Series Forecasting

General Objective

To critically analyze and present a Machine Learning (ML) algorithm or Deep Learning (DL) architecture for time series forecasting through a graduate-level technical presentation delivered entirely in English, demonstrating a rigorous understanding of its mathematical foundations, computational principles, and practical applications.

Project Guidelines

- **Format:** Each student will have a 25-minute technical presentation and a 10-minute question-and-answer session. Slides must be .pptx or .pdf, maintaining an appropriate balance between text and visual content. Students are encouraged to incorporate innovative scientific illustrations, informative diagrams, high-quality graphics, and relevant technical information to enhance clarity and audience engagement.
- **Animated and Interactive Visual Resources:** The presentation must include at least one animated visual resource, such as an animation, video, time-lapse sequence, or dynamic simulation. This resource should be displayed while the presenter actively explains the underlying concept, mathematical process, or algorithmic behavior.
- **Implementation:** Students must demonstrate a practical application of their assigned ML algorithm or DL architecture using an available dataset. This section must include an explanation of the main components of the implementation, a live execution of the code, and interpretation of the forecasting results. It is optional to compare the results with another baseline algorithms to present forecasting metrics like SSE, MSE, RMSE, MAE or R^2.
- **Literature Review:** Students must select and reference at least one scientific research article directly related to their assigned algorithm or architecture. After implementation, they must display the article and briefly explain its research objective, methodology, and principal findings, emphasizing its relevance to time series forecasting.
- **Conclusions:** The presentation must conclude with a clear and well-structured synthesis of the most important concepts, findings, and practical implications. Students should critically discuss the strengths, limitations, and potential applications of their assigned technique. The conclusions should encourage scientific curiosity, stimulate discussion, and motivate the audience to formulate meaningful questions during the question-and-answer session.

Topics

1. Decision Trees (DT).
2. Random Forest (RF).
3. Gradient Boosting Decision Trees (GBDT).
4. Recurrent Neural Networks (RNN).
5. Long Short-Term Memory (LSTM).
6. Gated Recurrent Unit (GRU).

Project Deliverables

The resources will be sent by email to the professor.

1. Technical presentation (.pptx or .pdf).
2. Source Code (.py or .ipynb)
3. Dataset (.csv or .xls)
4. Scientific Research Article (.pdf)
5. Conclusions Infographic (.pdf)

Project Presentation Schedule

Topic	Date
1. Decision Trees (DT).	November 9th, 2026
2. Random Forest (RF).	November 9th, 2026
3. Gradient Boosting Decision Trees (GBDT).	November 11th, 2026
4. Recurrent Neural Networks (RNN).	November 11th, 2026
5. Long Short-Term Memory (LSTM).	November 18th, 2026
6. Gated Recurrent Unit (GRU).	November 18th, 2026

Note: Presentation dates may change due to school activities or unexpected circumstances. Any changes will be announced in advance by email. Any doubt contact your professor.